PolymerFFKit
User Manual

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1 Introduction

PolymerFFKit is an interactive toolkit for polymer molecular dynamics workflows. It supports trajectory mapping, geometry extraction, Boltzmann inversion, bonded parameter fitting, non-bonded parameter assignment, polymer sequence generation, and topology export for GROMACS and LAMMPS. The current release is a terminal-based interactive application and supports AA, UA, and CG modeling workflows.

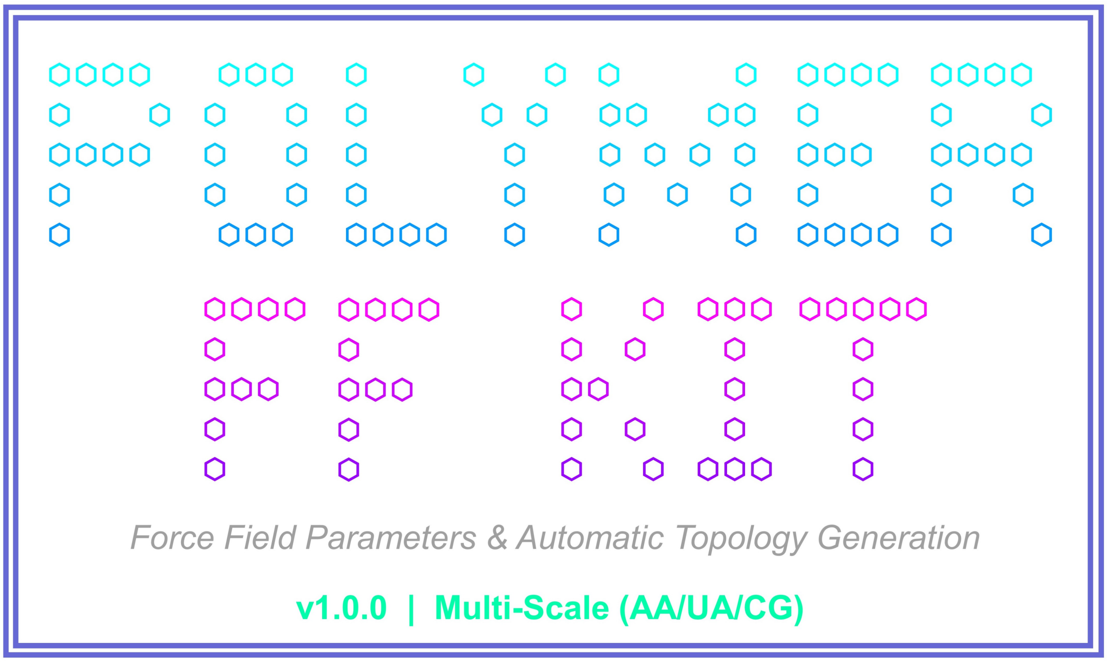

Figure 1: PolymerFFKit main page and workflow

2 System Requirements

Item	Requirement
Operating system	Windows 10/11 x64, Ubuntu 22.04+ x64, macOS 13+ Intel or Apple Silicon
Processor	Intel/AMD x86-64 or Apple Silicon
Memory	16 GB RAM or above recommended
Storage	512 GB SSD or above recommended; output directory must be writable
Optional tools	GROMACS, LAMMPS, VMD, or PyMOL for validation and visualization

3 Installation and Launch

3.1 Windows

Download `PolymerFFKit-v1.0.0-windows-x64.zip`, unzip it, and keep `PolymerFFKit.exe`, `data`, `templates`, and the manuals together. Double-click the executable or run:

```
.\PolymerFFKit.exe
```

3.2 Linux

Download `PolymerFFKit-v1.0.0-linux-x64.tar.gz`, extract it, and run:

```
tar -xzf PolymerFFKit-v1.0.0-linux-x64.tar.gz
cd PolymerFFKit
chmod +x PolymerFFKit
./PolymerFFKit
```

3.3 macOS

Download the Intel or Apple Silicon package, unzip it, and open `PolymerFFKit.app`. If macOS blocks the first launch, open System Settings, go to Privacy and Security, and allow the signed or notarized application.

4 Main Interface

The application starts with the main menu:

```
1  Trajectory Analysis & Force Field Generation
2  Polymer Sequence & Topology Builder
0  Exit gracefully
```

General interaction rules:

- Enter the menu number and press Enter.
- File path prompts support Tab completion and drag-and-drop paths.
- Press Enter to accept a displayed default value.
- Select `0` to exit from the main menu and `-1` to return from a submenu.
- Press Ctrl+C to interrupt the program safely.

5 Module 1: Trajectory Analysis and Force Field Generation

This module provides a four-step workflow for deriving force-field parameters from molecular dynamics trajectories.

5.1 Step 11: Trajectory Mapping and Extraction

Purpose: read a multi-frame XYZ trajectory and export an AA extracted, UA mapped, or CG mapped trajectory.

Main inputs:

- Input trajectory: `.xyz`
- Processing percentage, for example `80` for the last 80 percent of frames
- Custom UA or CG mapping rules: `mapping_rules.json`
- Center definition: COM or COG

Mapping rule example:

```
{
  "custom_masses": {
    "C1": 12.011,
    "H1": 1.008
  },
  "rules": [
```

```
    {"name": "A", "atoms": [1, 2, 3]},
    {"name": "B", "atoms": [4, 5, 6]}
  ]
}
```

Outputs include `AA_extracted.xyz`, `UA_mapped.xyz`, or `CG_mapped.xyz`.

5.2 Step 12: Geometry Extraction

Purpose: identify bonds, angles, and dihedrals from a mapped trajectory and export raw measured values.

Main inputs:

- Input trajectory: `.xyz`
- Output path: default `./raw_analysis.txt`
- Bond cutoff: about 0.8 nm for CG and about 0.2 nm for AA

Output example:

```
Raw Analysis Data Dump (Grouped by Type)
BOND_A-B: 0.3421,0.3510,0.3488
ANGLE_A-B-C: 112.4000,114.2000,116.0000
DIHEDRAL_A-B-C-D: -178.0000,60.1000,179.3000
```

Note: Step 12 writes raw text values. Step 13 expects binned histogram CSV input, so users usually convert raw values into histogram columns before fitting.

5.3 Step 13: Boltzmann Inversion and Fitting

Purpose: read geometry distribution CSV data, compute PMF curves, and fit bonded force-field parameters.

CSV rule: each data column must be placed immediately after its bin-coordinate column. Data columns must begin with `bond_`, `angle_`, or `dihedral_`.

```
r_nm,bond_A-B,theta_deg,angle_A-B-C,phi_deg,dihedral_A-B-C-D
0.280,12,90,2,-180,1
0.300,45,100,8,-120,5
0.320,80,110,30,-60,18
```

Main settings:

- Temperature: default 300 K
- PMF threshold: default 50 percent for bonds and angles, 100 percent for dihedrals
- Dihedral fitting: skip, periodic Fourier order 1-4, or Ryckaert-Bellemans
- Optional PNG PMF fitting plots

The default output directory is `./fit_results`, containing `forcefield_params.txt` and fit plots.

5.4 Step 14: Non-Bonded Parameter Assignment

Purpose: combine mapping units, fitted bonded parameters, built-in or custom ITP libraries, and optional external charges into a unified force-field JSON file.

Main inputs:

- Mapping rules JSON from Step 11
- `forcefield_params.txt` from Step 13
- Built-in or custom `.itp` force-field library

- Optional external charge file

Charge file example:

```
A 0.000
B -0.100
C 0.100
```

The default output is `ff_unified_config.json`. This file is used by Step 22. For complex topology contexts, extend it with `atom_naming`, `branch_definitions`, or `rigid_definitions` using the supplied template.

6 Module 2: Polymer Sequence and Topology Builder

6.1 Step 21: Polymer Sequence Generation

Purpose: generate randomized polymer sequences from repeat-unit types, ratios, and degree of polymerization.

Configuration example:

```
{
  "seed": 42,
  "number_of_sequences": 3,
  "strict": true,
  "residue": ["A", "B", "C"],
  "ratios": [5.0, 3.0, 2.0],
  "degree of aggregation": 100
}
```

With the default prefix `./output/polymer`, outputs are:

```
polymer_sequences.txt
polymer_analysis.txt
polymer_config.json
```

6.2 Step 22: 3D Structure and Topology Building

Purpose: convert one-dimensional sequences into 3D structures and topology files.

Main inputs:

- Sequence file from Step 21
- Unified force-field JSON from Step 14
- Non-bonded parameter ITP file
- Output directory, default `./topology_out`
- Number of molecules for the GROMACS `.top` file

Sequence example:

```
>Sequence_1|Len:12
AABACBBCAABC
>Sequence_2|Len:12
BBCACAAABBCA
```

Outputs:

```
CHAIN_1.pdb
CHAIN_1.data
CHAIN_1.itp
CHAIN_1.top
```

7 Input Templates

The `templates` directory contains editable examples for all common input formats:

- `xyz_trajectory_template.xyz`
- `mapping_rules_template.json`
- `fitting_histogram_csv_template.csv`
- `fitted_params_template.txt`
- `external_charges_template.txt`
- `sequence_config_template.json`
- `sequence_file_template.txt`
- `unified_config_template.json`
- `gromacs_itp_template.itp`

8 Recommended Workflow

1. Prepare an AA trajectory and mapping rules.
2. Run Step 11 to create the target-resolution trajectory.
3. Run Step 12 to extract raw bond, angle, and dihedral values.
4. Convert raw values into binned histogram CSV data.
5. Run Step 13 to fit bonded parameters.
6. Run Step 14 to assign non-bonded parameters and export a unified config.
7. Check or extend `atom_naming`.
8. Run Step 21 to generate polymer sequences.
9. Run Step 22 to generate PDB, ITP, TOP, and DATA files.

9 FAQ

The program cannot find built-in force-field libraries.
Check that the release package contains `data/ff_libraries` and that the directory stays inside the same package as the executable.

Step 13 does not fit a parameter.
Check column names, bin columns, number of valid points, and the PMF threshold. Data columns must begin with `bond_`, `angle_`, or `dihedral_`.

Step 22 generates no atoms.
Check that sequence symbols match `atom_properties` or `atom_naming`. Complex residue contexts require corresponding naming keys in the unified config.

Can outputs be used directly in simulation?
Outputs should be validated with GROMACS, LAMMPS, or visualization tools before production simulation.

10 License, Disclaimer, and Contact

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